Segre symbols

Pieter Belmans

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Abstract

The classification of (possibly singular) intersections of quadric hypersurfaces turns out to be completely classical, and it is the subject of the PhD thesis of Corrado Segre (of Segre embedding fame, not to be confused with his student Beniamino Segre). In this short note I will recall the definition, and introduce some of the competing terminology. I will also summarise the classification for $\mathbb{P}^2$ and $\mathbb{P}^3$.

1 Definition

Geometry Throughout we work of an algebraically closed field k of characteristic not two.

Consider a pencil in $\mathcal{O}_{\mathbb{P}^n}(2)$, i.e. we are considering a one-dimensional family of quadric hypersurfaces in $\mathbb{P}^n$. We are interested in classifying complete intersections of quadric hypersurfaces, which corresponds to classifying the possible base loci of pencils of quadrics, as any two (distinct) quadric hypersurfaces span a pencil and every such pencil describes a complete intersection of quadric hypersurfaces as its base locus.

By the correspondence between quadratic forms and bilinear forms, we can consider every element to be given by a symmetric $(n+1) \times (n+1)$-matrix. Let A and B be linearly independent elements in the pencil. Then every nonzero element of the pencil is described by the symmetric matrix $\mu A + \lambda B$, for $[\mu : \lambda] \in \mathbb{P}^1$. We can consider $\det(\mu A + \lambda B) \in k[\mu, \lambda]$, a homogeneous polynomial of degree $n+1$, defining at most $n+1$ points on $\mathbb{P}^1$.

From now on we will assume that B is has rank $n+1$, so the quadric hypersurface associated to B is smooth. This implies that $\det(\mu A+\lambda B) \neq 0$ at the point $[0:1]$, so we can effectively reduce our computations to the affine chart given by $[1:\lambda]$. This will indeed be the case for us, as we are considering divisors of bidegree $(2,2)$ on $\mathbb{P}^1 \times \mathbb{P}^1$, hence we will take B to correspond to $\mathbb{P}^1 \times \mathbb{P}^1$ embedded in $\mathbb{P}^3$ via the Segre embedding.

Linear algebra We are considering the polynomial $\det(A+\lambda B)$ of degree $n+1$. Let $\alpha_1, \ldots, \alpha_r$ be its (distinct) roots. The elements in the linear system corresponding to these roots are precisely the singular quadric hypersurfaces in the pencil. We will be interested in understanding their structure.

For a root α_i of multiplicity e_i we know that it the singular quadric hypersurface associated to it is given by a cone over a smooth quadric. If α_i is not only a zero of $\det(A+\lambda B)$, but also of all its subdeterminants of size $n-h_i+2$, for $h_i \geq 2$ (where we choose h_i maximal in this fashion), then the dimension of the vertex of the cone is precisely $h_i - 1$ (so the case where $h_i = 1$ corresponds to the generic situation, where the vertex is just a point).

Then define l^i_j to be the minimum multiplicity of the root α_i for the set of subdeterminants of size $n+1-i$, for $i = 0, 1, \ldots, h_i - 1$. Then $l^i_j \geq l^{i+1}_j$, and we define $e^i_j := l^i_j - l^{i+1}_j$ for $j = 0, \ldots, h_i - 1$, such that $e^i_j = \sum_{j=0}^{h_i-1} e^i_j$. It can be shown that $e^i_j \leq e^i_{j+1}$. All this gives us a factorisation

$$\det(A+\lambda B) = \prod_{j=0}^{h_i-1} (\lambda - \alpha_i)^{e^i_j} f_i(\lambda) \tag{1}$$

where $f_i(\alpha_i) \neq 0$.

The main definition We now define the following.

Definition 1. The polynomials $(\lambda - \alpha_i)^{e^i_j}$ are the *elementary divisors* associated to α_i, and the exponents e_i are the *characteristic numbers*.

Remark 2. All of this can be done over an arbitrary field for an element $M(\lambda)$ of $\mathrm{Mat}_n(k[\lambda])$. Applying this to $M(A+\lambda 1_n)$, for $A \in \mathrm{Mat}_n(k)$ we see that the product of the elementary divisors is the characteristic polynomial of A, the least common multiple of the elementary divisors is the minimum polynomial of A.

Also, if k is a splitting field of the characteristic polynomial, then there is a connection to the Jordan normal form. Indeed, the number of elementary divisors corresponds to the number of Jordan cells, etc.

We are now ready to define Segre symbols.

Definition 3. The *Segre symbol* of the pencil is

$$[(e^0_0, \ldots, e^0_{h_0-1}); (e^1_0, \ldots, e^1_{h_1-1}); \ldots; (e^r_0, \ldots, e^r_{h_r-1})]. \tag{2}$$

One can then show the following result.

Theorem 4 (Weierstrass, Segre)**.** Consider two pencils of quadric hypersurfaces in $\mathbb{P}^n$. Then their base loci are projectively equivalent if and only if they have the same Segre symbol and there exists an automorphism of $\mathbb{P}^1$ taking $(1 : \alpha_i)$ to $(1 : \beta_i)$.

Remark 5. The number of Segre symbols for the intersection of two quadrics in a fixed dimension is given by the OEIS sequence A001970.

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2 Examples

2.1 Pencils of quadrics in $\mathbb{P}^2$

In table 1 we give the Segre symbols for pencils of quadrics in $\mathbb{P}^2$. For more details one is referred to [3, pages 304–305].

Segre symbol	description
$[1,1,1]$	four distinct points
$[(1,1),1]$	two double points
$[2,1]$	a double point and two other points
$[(2,1)]$	quadruple point
$[3]$	a triple point and another point

Table 1: Segre symbols for pencils of quadrics in $\mathbb{P}^2$

2.2 Pencils of quadrics in $\mathbb{P}^3$

In table 2 we give the Segre symbols for pencils of quadrics in $\mathbb{P}^3$. For more details one is referred to [3, pages 305–308], or for even more details to [4, §13.8].

Segre symbol	divisor
$[1,1,1,1]$	elliptic curve
$[3,1]$	cuspidal curve
$[2,1,1]$	nodal curve
$[(1,1),1,1]$	two conics intersecting in two points
$[(2,1),1]$	two tangent conics
$[2,(1,1)]$	a conic and two lines in a triangle
$[(3,1)]$	a conic and two lines intersecting in one point
$[(1,1),(1,1)]$	quadrangle
$[2,2]$	twisted cubic and a bisecant
$[4]$	twisted cubic and a tangent line
$[(1,1,1),1]$	double conic
$[(2,1,1)]$	two double lines
$[(2,2)]$	a double line and two lines in general position

Table 2: Segre symbols for pencils of quadrics in $\mathbb{P}^3$

2.3 Pencils of quadrics in $\mathbb{P}^4$

This turns out to be the classification of (possibly singular) del Pezzo surfaces of degree 4 (also called Segre quartic surfaces), and is discussed in [1, §8.6].

A Segre symbols for pencils without smooth quadric (by Thilo Baumann)

In general, the pencil of quadrics $\mu A + \lambda B$ does not need to contain a smooth quadric. Instead, due to [2, Chapter VIII, §10, Theorem III] we can bring the pair of quadrics defined by A and B simultaneously into a normal form. To do so, let $r \in \mathbb{Z}_{>0}$ and define the homogeneous polynomials

$$q_r^+(x_0, \ldots, x_{2r}) = \sum_{i=1}^{r} 2x_{2i-2}x_{2i-1} \quad \text{and} \quad q_r^-(x_0, \ldots, x_{2r}) = \sum_{i=1}^{r} 2x_{2i-1}x_{2i}. \tag{3}$$

The theorem says that one can find coordinates

$$x_0^{(0)}, \ldots, x_{r_0-1}^{(0)}, x_0^{(1)}, \ldots, x_{2r_1}^{(1)}, \ldots, x_0^{(k)}, \ldots, x_{2r_k}^{(k)}, z_0, \ldots, z_m \tag{4}$$

of $\mathbb{P}^n$ such that

$$n + 1 = r_0 + \sum_{i=1}^{k} (2r_i + 1) + m. \tag{5}$$

Then the quadric Q_A defined by A is

$$Q_A = \mathbb{V}\left(q_{r_1}^+(x_0^{(1)}, \ldots, x_{2r_1}^{(1)}) + \ldots + q_{r_k}^+(x_0^{(k)}, \ldots, x_{2r_k}^{(k)}) + q_A(z_1, \ldots, z_m)\right) \subseteq \mathbb{P}^n, \tag{6}$$

and the quadric Q_B defined by B is

$$Q_B = \mathbb{V}\left(q_{r_1}^-(x_0^{(1)}, \ldots, x_{2r_1}^{(1)}) + \ldots + q_{r_k}^-(x_0^{(k)}, \ldots, x_{2r_k}^{(k)}) + q_B(z_1, \ldots, z_m)\right) \subseteq \mathbb{P}^n. \tag{7}$$

Moreover the quadric $\widetilde{Q}_B = \mathbb{V}(q_B) \subset \mathbb{P}^{m-1}$ is smooth.

Combining the theorem with the construction of Segre symbol $[\underline{e}^0, \ldots, \underline{e}^r]$ for the induced pencil in $\mathbb{P}^m$, one defines the Segre symbol of the pencil $\mu A + \lambda B$ to be

$$[\underline{e}^0, \ldots, \underline{e}^r; r_1, \ldots, r_k; r_0] \tag{8}$$

Example 6. Following [5, Table 0], there are three more pencils of quadrics in $\mathbb{P}^2$ consisting of only singular quadrics.

Segre symbol	Normal form	m	Discriminant locus for pencil in $\mathbb{P}^m$
$[;1;]$	$Q_A = \mathbb{V}(x_0x_1)$, $Q_B = \mathbb{V}(x_1x_2)$	0	-
$[1,1;;1]$	$Q_A = \mathbb{V}(x_0^2)$, $Q_B = \mathbb{V}(x_0^2 + x_1^2)$	2	two distinct points
$[2;;1]$	$Q_A = \mathbb{V}(x_0^2)$, $Q_B = \mathbb{V}(x_0x_1)$	2	one double point

Table 3: Segre symbols for pencils of quadrics in $\mathbb{P}^2$.

References

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